SHAGUFTA ANJUM

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EDUCATION

University of Illinois Urbana-Champaign Illinois, USA | Jan 2024 - April 2025
Master of Computer Science ***GPA: 4/4***
Coursework: Distributed Systems, Cloud Computing, Applied ML, Data Mining, LLMs, Computer Vision, Human Computer Interaction

Mahindra University India | July 2019 - May 2023
Bachelor of Technology, Computer Science Engineering ***GPA: 3.95/4***
Coursework: Operating Systems, Distributed Systems, Computer Architecture, Algorithms, Numerical Methods, Probability & Statistics, Networks

EXPERIENCE

Cognida.ai Sunnyvale, CA, USA
Software Engineer *Jan 2026 – present*
- Built large-scale data ingestion and preprocessing pipelines for ERP financial datasets, enabling mapping across heterogeneous systems
- Implemented NLP-based entity matching with batch processing, improving throughput and reducing manual reconciliation overhead
- Developed backend services for high-volume API ingestion, transformation, and export under strict reliability and performance constraints

Amazon New York, NY, USA
Software Engineer *June 2025 – Dec 2025*
- Implemented scalable backend services for bulk user operations across 10K+ enterprise accounts, supporting high-volume batch workflows
- Designed graph algorithms for hierarchical account relationships with Redis adjacency caching, reducing authorization latency by ~35%
- Parallelized compute-intensive workflows and optimized concurrency handling, reducing processing time by 40–50% under peak load
- Built distributed APIs with complex permission evaluation logic, improving system throughput and operational reliability
- Integrated runtime configuration and feature-gating systems to support safe deployment under production traffic

GlobalLogic Inc. Santa Clara, CA, USA
Software Engineer Intern *June 2024 – July 2024*
- Architected a real-time multiplayer platform using event-driven services and WebSocket coordination for concurrent user interactions
- Designed backend services supporting high-frequency state updates, caching, and role-based access control supporting thousands of users
- Implemented Redis caching for frequently accessed session data to reduce database load and improve response latency

Dell Technologies Bengaluru, India
Software Engineer *July 2023 – Dec 2023*
- Built a real-time computer vision inference pipeline for manufacturing defect detection meeting strict processing latency constraints
- Architected a Kafka event-driven ingestion layer to stream image metadata and inference payloads for scalable, fault-tolerant processing
- Implemented backend services to orchestrate model inference, manage task scheduling, and coordinate asynchronous result publishing
- Optimized end-to-end pipeline performance by reducing serialization overhead and batching inference requests

PROJECTS

Personalized Writing Assistant using Fine-tuned LLMs *Sep 2024 – Dec 2024*
- Built a RAG inference pipeline using vector indexing and dynamic context injection to reduce hallucination in generation workflows
- Fine-tuned language models using parameter-efficient methods on domain-specific datasets while keeping compute & memory overhead low

VisionAid – Smart Assistance for the Visual Impaired *June 2023 – Nov 2023*
- Developed real-time object detection pipelines using computer vision models and streaming inputs to support low-latency detection
- Designed a voice-driven interaction layer to convert detection outputs into audio navigation cues, enabling accessible user guidance
- Reduced end-to-end inference latency by optimizing frame batching, model execution paths, and stream I/O

SKILLS

- Programming Languages: Python, Java, JavaScript, Typescript
- Systems and Backend: Distributed Systems, Concurrency, Caching strategies, Event-driven architecture, REST APIs
- Data and Infrastructure: MongoDB, MySQL, PostgreSQL, Redis, Kafka
- Machine learning: PyTorch, TensorFlow, Scikit-learn, Pandas, Huggingface
- Cloud and Tooling: AWS, GCP, Docker, Kubernetes, CI/CD, GitHub Actions